National University of Singapore
School of Computing
CS1101S: Programming Methodology
Semester I, 2026/2027

R4
Data Abstraction

Python §2

1. `pair(x, y)`: makes a pair from `x` and `y`
2. `head(p)`: returns the head (first component) of the pair `p`
3. `tail(p)`: returns the tail (second component) of the pair `p`
4. `llist(x1, x2, x3, ..., xn)`: returns a llist with `n` elements. The first element is `x1`, the second `x2`, etc.
5. `len(xs)`: returns the length of the llist `xs`
6. `llist_ref(xs, n)`: returns element of llist `xs` at position `n`, where the first element is at position 0.

Problems:

1. Draw the box-and-pointer diagrams for the results of evaluating the following programs. Also write the data structures in box notation and in llist notation.
   (a) `pair(1, 2)`
   (b) `pair(1, pair(3, pair(5, None)))`
   (c) `pair(pair(pair(3, 2), pair(1, 0)), None)`
   (d) `pair(0, llist(1, 2))`
2. Write Python §2 programs that print the following:
   (a) `[1, [2, [3, None]]]`
   (b) `[1, [2, 3]]`
   (c) `[[1, [2, None]], [[3, [4, None]], [[5, [6, None]], None]]]`
3. Write programs that only use applications of `print`, `head` and `tail` and the name `lst` so that the value 4 is printed:
   (a) `lst = llist(7, 6, 5, 4, 3, 2, 1)`
   (b) `lst = llist(llist(7), llist(6, 5, 4), llist(3, 2), 1)`
   (c) `lst = llist(7, llist(6, llist(5, llist(4, llist(3, llist(2, llist(1)))))))`

   Note: The key skill for this question is to translate an expression into a box-and-pointer diagram and to systematically traverse the data structure.

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